Pen Pals

Muwahid

MTH631 - Quiz 2

MTH631:Quiz-2

Quiz Start Time: 09:59 AM, 18 February 2021

Question # 5 of 10 (Start time: 09:59:29 AM, 18 February 2021)

Total Marks: 1

A compact set in $\mathbb{R}^n$ is - - - - - - - - - - .

Select the correct option

Reload Math Equations

○	open and unbounded
○	open and bounded
○	closed and unbounded
○	closed and bounded



MTH631:Quiz-2

Quiz Start Time: 09:59 AM, 18 February 2021

Question # 7 of 10 (Start time: 10:00:13 AM, 18 February 2021)

Total Marks: 1

In $\mathbb{R}^n$, an open n - ball of radius r about X_0 is $B_r(X_0) = \{X : |X - X_0| < r\}$, then its closure is - - - - - - - .

Select the correct option

Reload Math Equations

○	$\{X : \|X - X_0\| = r\}$
○	$\{X : \|X - X_0\| \leqslant r\}$
○	$\{X : \|X - X_0\| \geqslant r\}$
○	$\{X : \|X - X_0\| > r\}$

MTH631:Quiz-2

Quiz Start Time: 09:59 AM, 18 February 2021

Question # 10 of 10 (Start time: 10:01:14 AM, 18 February 2021)

Total Marks: 1

In $\mathbb{R}^2$, the set $\left\{(x,y) : \left(x^2+y^2 \leqslant a\right) \vee \left(x^2+y^2 \geqslant b\right), a < b\right\}$ is a region.

Select the correct option

Reload Math Equations

- True
- False

MTH631:Quiz-2

Quiz Start Time: 09:59 AM, 18 February 2021

Question # 9 of 10 (Start time: 10:00:44 AM, 18 February 2021)

Total Marks: 1

If $\mathbb{R}^n$ is connected, such that $\mathbb{R}^n = A \cup B$ with $\bar{A} \cap B = A \cap \bar{B} = \phi$, then - - - - - .

Select the correct option

Reload Math Equations

- $\bar{A} \subset A, \bar{B} \subset B$
- A and B are both open and closed
- Either $A = \mathbb{R}^n, B = \phi$ or $B = \mathbb{R}^n, A = \phi$
- All above can be concluded

MTH631:Quiz-2

Question # 6 of 10 (Start time: 09:56:07 AM, 18 February 2021)

If $\lim_{X \to X_0} f(X)$ exists, then it is _____.

Select the correct option

○	1
◉	unique
○	0
○	none of these

MTH631:Quiz-2

Question # 3 of 10 (Start time: 09:54:16 AM, 18 February 2021)

Sequence in $\mathbb{R}^2$ defined as;

$\{X_r\} = \left\{\left(\frac{\cos r}{r}, \frac{\sin r}{r}\right)\right\}$ converges to - - - - - .

Select the correct option

	$(1, 0)$
	$(0, 1)$
	$(0, 0)$
	$(1, 1)$

Question # 8 of 10 (Start time: 09:56:56 AM, 18 February 2021)

An open disc: $\{(x, y) : x^2 + y^2 < 1\}$ in $\mathbb{R}^2$ is - - - - - - .

Select the correct option

- [x] diconnected
- [ ] disconnected polygonally
- [ ] connected polygonally
- [ ] none of these.

Question # 10 of 10 (Start time: 09:57:27 AM, 18 February 2021)

In $\mathbb{R}^n$, if the line segment: $X = tX_2 + (1-t)X_1$, $0 < t < 1$, joining X_1 and X_2 lies in $S_\varepsilon(X_0)$, *then* $----$.

Select the correct option

- [x] $|X - X_0| < \varepsilon$
- [ ] $|X - X_0| < 1 - \varepsilon$
- [ ] $|X - X_0| < t$
- [ ] $|X - X_0| < 1 - t$

Question # 4 of 10 (Start time: 09:54:46 AM, 18 February 2021)

Set of isolated point(s) of the complement of set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $- - - - -$.

Select the correct option

- $\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\} \cup \{(0,0)\}$
- $\{(0,0)\}$
- $\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\} \cap \{(0,0)\}$
- $\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\}$

Question # 2 of 10 (Start time: 09:53:49 AM, 18 February 2021)

In $\mathbb{R}^3$, the lines $L_1 : X = (2, -1, 5) + \alpha(2, -1, 3)$ and $L_2 : X = (2, -1, 5) + \beta\left(-5, \frac{5}{2}, -\frac{15}{2}\right)$ are traversed in - - - - - - - - - directions, where $-\infty < \alpha, \beta < \infty$.

Select the correct option

- [x] same
- [ ] opposite
- [ ] perpendicular
- [ ] oblique

Question # 5 of 10 (Start time: 09:55:47 AM, 18 February 2021)

The function $f(x, y) = \frac{xy}{x^2+y^2}$ is defined everywhere in _____ except at _____.

Select the correct option

○	$R^2,\ (1, -1)$
○	$R^3,\ (1, -1, 0)$
●	$R^2,\ (0, 0)$
○	none of these

MTH631:Quiz-2

Question # 7 of 10 (Start time: 09:56:28 AM, 18 February 2021)

If $\phi \neq S \subseteq \mathbb{R}^n$, then the set S is bounded if - - - - - - - .

Select the correct option

- $\sup\{|X-Y| : X, Y \in S\} = \infty$
- $\sup\{|X-Y| : X, Y \in S\} < \infty$
- $\inf\{|X-Y| : X, Y \in S\} = \infty$
- $\inf\{|X-Y| : X, Y \in S\} < \infty$

Question # 9 of 10 (Start time: 09:57:13 AM, 18 February 2021)

The space R^n is _____.

Select the correct option

- (●) connected
- ( ) not connected
- ( ) both a and b
- ( ) none of these

Question # 1 of 10 (Start time: 09:53:22 AM, 18 February 2021)

For a non empty closed and bounded subset S in $\mathbb{R}^n$, if H is the collection open sets such that $S \subset \cup\{H : H \in \mathrm{H}\}$, then by Heine - Borel theorem,

Select the correct option

○	$S \subset \underset{\lambda\in\Lambda}{\cup}\{H_\lambda : H_\lambda \in \mathrm{H}\}$
○	$S \subset \overset{\infty}{\underset{\alpha=1}{\cup}}\{H_\alpha : H_\alpha \in \mathrm{H}\}$
○	$S \subset \overset{n}{\underset{\alpha=1}{\cup}}\{H_\alpha : H_\alpha \in \mathrm{H}\}$
●	All above are equally valid

MTH631:Quiz-2

Quiz Start Time: 09:40 AM, 18 February 2021

Question # 7 of 10 (Start time: 09:44:01 AM, 18 February 2021)

Total Marks: 1

Limit points of set $S = \{(x, y) : -n < x, y < n, (x, y) \neq (0, 0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, – – – –.

Select the correct option

Reload Math Equations

- $\{(0, 0)\} \cup \{(x, y) : x, y = \pm n, n \in \mathbb{N}\}$
- $\{(0, 0)\} \cup S$
- $\{(0, 0)\} \cup \{(x, y) : x, y = \pm n, n \in \mathbb{N}\} \cup S$
- $\{(x, y) : x, y = \pm n, n \in \mathbb{N}\} \cup S$

MTH631:Quiz-2

Quiz Start Time: 09:40 AM, 18 February 2021

Question # 6 of 10 (Start time: 09:43:25 AM, 18 February 2021)

Total Marks: 1

If $\lim_{X \to X_0} f(X)$ exists, then it is _____.

Select the correct option

Reload Math Equations

- 1
- unique
- 0
- none of these

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 09:42:19 AM, 18 February 2021) Total Marks: 1

Closure of the set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $-----$.

Select the correct option

Reload Math Equations

○	$\{(x,y) : -n < x, y < n, n \in \mathbb{N}\}$
○	$\{(x,y) : -n \leqslant x, y \leqslant n, n \in \mathbb{N}\}$
○	$\{(x,y) : (x,y) \neq (0,0)\}$
○	$\{(x,y) : x, y \leqslant n, (x,y) \neq (0,0), n \in \mathbb{N}\}$



Question # 3 of 10 (Start time: 09:41:27 AM, 18 February 2021) Total Marks: 1

Exterior of the set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $----$.

Select the correct option

Reload Math Equations

- $\{(x,y) \in \mathbb{R}^2 | \, |x| < n, |y| > n, n \in \mathbb{N}\}$
- $\{(x,y) \in \mathbb{R}^2 | \, |x| > n, |y| < n, n \in \mathbb{N}\}$
- $\{(x,y) \in \mathbb{R}^2 | \, |x| > n, |y| > n, n \in \mathbb{N}\}$ ✓
- $\{(x,y) \in \mathbb{R}^2 | \, |x| < n, |y| < n, n \in \mathbb{N}\}$

Click to Save Answer & Move to Next Question

2021)

In $\mathbb{R}^n$, an open n - ball of radius r about X_0 is $B_r(X_0) = \{X : |X - X_0| < r\}$, which contains - - - - - - - .

an ε - neighborhood of each of its points

limit points of each of its points

only its isolated point at center X_0

limit points of each of its boundary

Question # 2 of 10 (Start time: 09:04:48 AM, 18 February 2021) Total Marks: 1

Limit points of set $S = \{(x, y) : -n < x, y < n, (x, y) \neq (0, 0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, - - - - -.

Select the correct option

Reload Math Equations

- $\{(0, 0)\} \cup \{(x, y) : x, y = \pm n, n \in \mathbb{N}\}$
- $\{(0, 0)\} \cup S$
- $\{(0, 0)\} \cup \{(x, y) : x, y = \pm n, n \in \mathbb{N}\} \cup S$
- $\{(x, y) : x, y = \pm n, n \in \mathbb{N}\} \cup S$

MTH631:Quiz-2

Quiz Start Time: 09:40 AM, 18 February 2021

Question # 2 of 10 (Start time: 09:40:30 AM, 18 February 2021)

Total Marks: 1

Which of the following non - empty subset on Real line $\mathbb{R}$ is taken as *region* ?

Select the correct option

Reload Math Equations

- ○ Natural numbers $\mathbb{N}$
- ● Intervals (open, closed, semi open or closed)
- ○ Rationals $\mathbb{Q}$ or Irrationals $\mathbb{Q}^c$
- ○ Range of pointwise or uniform real valued convergent sequences

Click to Save Answer & Move to Next Question

MTH631:Quiz-2

Question # 10 of 10 (Start time: 09:43:48 PM, 17 February 2021)

The function $f(x,y) = \frac{xy}{x^2+y^2}$ is defined everywhere in _____ except at _____.

Select the correct option

○	$R^2,\ (1,-1)$
○	$R^3,\ (1,-1,0)$
○	$R^2,\ (0,0)$
○	none of these

In $\mathbb{R}^n$, if the line segment: $X = tX_2 + (1-t)X_1$, $0 < t < 1$, joining X_1 and X_2 lies in $S_\varepsilon(X_0)$, then

correct option

$|X - X_0| < \varepsilon$

$|X - X_0| < 1 - \varepsilon$

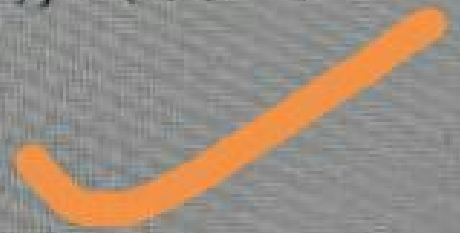
$|X - X_0| < t$

$|X - X_0| < 1 - t$

Intervals $(0,1)$ and $(1,2)$ are example of disconnected sets in $\mathbb{R}$ because - - - - - - - - .

$$(0,1) \cap (1,2) = \phi$$

$\{\text{closure of } (0,1)\} \cap (1,2) = \phi$ and $\{\text{closure of } (1,2)\} \cap (0,1) = \phi$



Click to Save Answer

MTH631:Quiz-2

Question # 7 of 10 (Start time: 09:41:51 PM, 17 February 2021)

Let $f(x,y)=\frac{xy}{x^2+y^2}$ then the limit of f along the line $y=-x$ as (x,y) approach $(0,0)$ is _____.

Select the correct option

○	0
○	undefined
○	$1/2$
○	$-1/2$

MTH631:Quiz-2

Question # 6 of 10 (Start time: 09:41:19 PM, 17 February 2021)

For a non empty closed and bounded subset S in $\mathbb{R}^n$, if H is the collection open sets such $S \subset \cup\{H : H \in \mathrm{H}\}$, then by Heine - Borel theorem,

Select the correct option

○	$S \subset \bigcup_{\lambda \in \Lambda} \{H_\lambda : H_\lambda \in \mathrm{H}\}$
○	$S \subset \bigcup_{\alpha=1}^{\infty} \{H_\alpha : H_\alpha \in \mathrm{H}\}$
○	$S \subset \bigcup_{\alpha=1}^{n} \{H_\alpha : H_\alpha \in \mathrm{H}\}$
○	All above are equally valid

Question # 5 of 10 (Start time: 09:40:39 PM, 17 February 2021)

An open disc: $\{(x, y) : x^2 + y^2 < 1\}$ in $\mathbb{R}^2$ is - - - - - - - .

Select the correct option

○	diconnected
○	disconnected polygonally
○	connected polygonally
○	none of these.

Exterior of the set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is – – – – –.

correct option

Re

$\{(x,y) \in \mathbb{R}^2 | |x| < n, |y| > n, n \in \mathbb{N}\}$

$\{(x,y) \in \mathbb{R}^2 | |x| > n, |y| < n, n \in \mathbb{N}\}$

$\{(x,y) \in \mathbb{R}^2 | |x| > n, |y| > n, n \in \mathbb{N}\}$



$\{(x,y) \in \mathbb{R}^2 | |x| < n, |y| < n, n \in \mathbb{N}\}$

Question # 3 of 10 (Start time: 09:39:20 PM, 17 February 2021)

A sequence of points $\{X_r\}$ in R^n converges if and only if for each _____ there is an integer K such that _____ if $r, s \geq K$.

Select the correct option

O	$\varepsilon < 0, \quad \lvert X_r - X_s \rvert < \varepsilon$
O	$\varepsilon > 0, \quad \lvert X_r - X_s \rvert > \varepsilon$
O	$\varepsilon > 0, \quad \lvert X_r - X_s \rvert < \varepsilon$
O	$\varepsilon \neq 0, \quad \lvert X_r - X_s \rvert > \varepsilon$

MTH631:Quiz-2

Question # 2 of 10 (Start time: 09:38:28 PM, 17 February 2021)

If $\lim_{X \to X_0} f(X)$ exists, then it is _____.

Select the correct option

○	1
○	unique
○	0
○	none of these

MTH631:Quiz-2

Question # 1 of 10 (Start time: 09:37:10 PM, 17 February 2021)

A compact set in R^n is - - - - - - - - - - .

Select the correct option

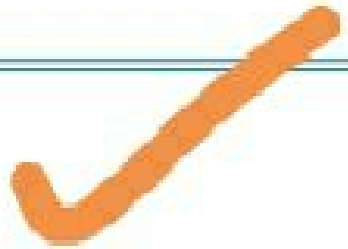
○	open and unbounded
○	open and bounded
○	closed and unbounded
○	closed and bounded

Closure of the set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $-----$.

Reload Math Equa

$\{(x,y) : -n < x, y < n, n \in \mathbb{N}\}$

$\{(x,y) : -n \leqslant x, y \leqslant n, n \in \mathbb{N}\}$



$\{(x,y) : (x,y) \neq (0,0)\}$

$\{(x,y) : x, y \leqslant n, (x,y) \neq (0,0), n \in \mathbb{N}\}$

Interior of the set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $- - - - -$.

ion

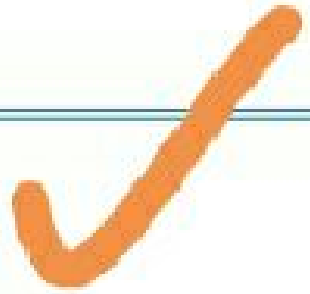
Reload Math Eq

$\{(x,y) \in \mathbb{R}^2 \,|\, |x| < n, |y| > n, n \in \mathbb{N}\}$

$\{(x,y) \in \mathbb{R}^2 \,|\, |x| > n, |y| < n, n \in \mathbb{N}\}$

$\{(x,y) \in \mathbb{R}^2 \,|\, |x| > n, |y| > n, n \in \mathbb{N}\}$

$\{(x,y) \in \mathbb{R}^2 \,|\, |x| < n, |y| < n, n \in \mathbb{N}\}$



Question # 9 of 10 (Start time: 09:31:00 PM, 17 February 2021)

If H is an open covering of a compact subset S, then S can be covered by _____ many sets from H.

Select the correct option

Reloa

○	finitely
○	infinitely
○	both a and b
○	non of these

Set of isolated point(s) of the complement of set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is – –

ct option

$\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\} \cup \{(0,0)\}$

$\{(0,0)\}$



$\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\} \cap \{(0,0)\}$

$\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\}$

Question # 9 of 10 (Start time: 09:29:43 PM, 17 February 20

The space R^n is _____.

Select the correct option

- ○ connected
- ○ not connected
- ○ both a and b
- ○ none of these

Question # 7 of 10 (Start time: 09:29:19 PM, 17 February 2021) Total Marks

A set " S " is polygonally connected if, ------ pair of points in S can be connected by a polygonal path lying ------- in " S ".

Select the correct option

- some, entirely
- every , entirely
- some, partially
- every, partially

Question # 8 of 10 (Start time: 09:29:10 PM, 17 February 2021)

If $\phi \neq .S \subseteq \mathbb{R}^n$, then the set S is bounded if - - - - - - - - .

Select the correct option

○	$\sup\{\|X-Y\| : X, Y \in S\} = \infty$
○	$\sup\{\|X-Y\| : X, Y \in S\} < \infty$ ✓
○	$\inf\{\|X-Y\| : X, Y \in S\} = \infty$
○	$\inf\{\|X-Y\| : X, Y \in S\} < \infty$

Question # 7 of 10 (Start time: 09:28:21 PM, 17 February 2021)

The function $f(x,y) = \frac{xy}{x^2+y^2}$ is defined everywhere in _____ except at _____.

Select the correct option

Reload Math

- $R^2,\ (1,-1)$
- $R^3,\ (1,-1,0)$
- $R^2,\ (0,0)$
- none of these

10 (Start time: 09:28:33 PM, 17 February 2021) Total Mark

In $\mathbb{R}^n$, which of the following is true about $\phi = \{\}$ and $A = \{(a_1, a_2, \ldots, a_n), a_i \in \mathbb{R}, 1 \leqslant i \leqslant n, i \in \mathbb{N}\}$?

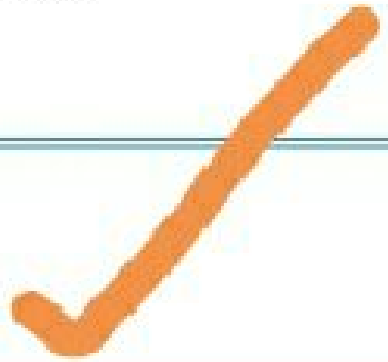
ect option

Reload Math Equatio

ϕ is Disconnected and A is Connected

ϕ is Connected and A is Disconnected

Both are Connected



Both are Disconnected

Question # 5 of 10 (Start time: 09:27:50 PM, 17 February 2021)

An open disc: $\{(x, y) : x^2 + y^2 < 1\}$ in $\mathbb{R}^2$ is - - - - - - .

Select the correct option

Relo

○	diconnected
○	disconnected polygonally
○	connected polygonally
○	none of these.

Tot

Boundary of the set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is - - - -

elect the correct option

Reload Math

○	$\{(x,y) \in \mathbb{R}^2 \mid x = \pm n, y = \pm n, n \in \mathbb{N}\} \cup \{(0,0)\}$
○	$\{(x,y) \in \mathbb{R}^2 \mid x = \pm n, y = \pm n, n \in \mathbb{N}\}$
○	$\{(0,0)\}$
○	$\{(x,y) \in \mathbb{R}^2 \mid x = \pm n, y = \pm n, n \in \mathbb{N}\} \cap \{(0,0)\}$

In $\mathbb{R}^n$, if the line segment: $X = tX_2 + (1-t)X_1$, $0 < t < 1$, joining X_1 and X_2 lies in $S_\varepsilon(X_0)$, *then* $----$.

he correct option

Reload Math Equatio

$|X - X_0| < \varepsilon$

$|X - X_0| < 1 - \varepsilon$

$|X - X_0| < t$

$|X - X_0| < 1 - t$

Question # 4 of 10 (Start time: 09:26:31 PM, 17 February 2021) **Total Marks: 1**

If H is an open covering of a compact subset S, then S can be covered by _____ many sets from H.

Select the correct option

Reload Math Equations

- finitely
- infinitely
- both a and b
- non of these

uestion # 2 of 10 (Start time: 09:25:16 PM, 17 February 2021) Total Marks: 1

Set of isolated point(s) of the complement of set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $----$.

elect the correct option

Reload Math Equations

- $\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\} \cup \{(0,0)\}$
- $\{(0,0)\}$
- $\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\} \cap \{(0,0)\}$
- $\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\}$

Question # 2 o 10 (Start time: 09:25:12 PM, 17 February 2021)

Total Marks: 1

A sequence c points $\{X_r\}$ in R^n converges if and only if for each _____ there is an integer K such that _____ if $r, s \geqslant K$..

Select the corr ct option

Reload Math Equations

- $\varepsilon <$, $|X_r - X_s| < \varepsilon$
- $\varepsilon >$, $|X_r - X_s| > \varepsilon$
- $\varepsilon >$, $|X_r - X_s| < \varepsilon$
- $\varepsilon \neq$, $|X_r - X_s| > \varepsilon$

Start time: 09:24:35 PM, 17 February 2021)

In $\mathbb{R}^n$, which of the following is true about $\phi = \{\}$ and $A = \{(a_1, a_2, \ldots, a_n), a_i \in \mathbb{R}, 1 \leqslant i \leqslant n, i \in N\}$?

ption

Reload M

ϕ is Disconnected and A is Connected

ϕ is Connected and A is Disconnected

Both are Connected



Both are Disconnected

Limit points of set $S = \{(x, y) : -n < x, y < n, (x, y) \neq (0, 0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, - - - --.

ion

Reload Ma

$\{(0, 0)\} \cup \{(x, y) : x, y = \pm n, n \in \mathbb{N}\}$

$\{(0, 0)\} \cup S$

$\{(0, 0)\} \cup \{(x, y) : x, y = \pm n, n \in \mathbb{N}\} \cup S$



$\{(x, y) : x, y = \pm n, n \in \mathbb{N}\} \cup S$

Click to Save Answer & Move to Next Que